

SWIFTSHIP TRACKER

Phase 1:

Requirement Analysis & Planning

Customers often face delays and confusion while booking, tracking, and managing parcel deliveries because they need to call support or use multiple apps. A conversational AI agent can simplify this by allowing customers to book parcels, get live tracking, and manage deliveries in one place.”

Key Requirements:

- Requirement gathering from end users and delivery agents.
- Defining system scope and user personas (Customer, Agent, Admin).
- Designing the data model with relationships between Parcel, Sender, Receiver, and Delivery.
- Identifying use cases such as parcel booking, tracking, and delivery update automation.
- Selecting tools: Salesforce Flows, Prompt Builder, and AgentForce AI.

Gathering & Analysing User Needs

Users Involved:

- Customers / Senders – End users who book parcels, track deliveries, and receive real-time updates through the portal or AI agent.
- Delivery Agents – Manage assigned parcels, update delivery status, and ensure timely handover to customers.
- Customer Support Team – Handle delivery-related issues, respond to customer queries, and ensure smooth communication.
- Admin Team – Configure the Salesforce environment, manage user roles, security settings, and monitor operational performance.
- System Integrator / Developer – Responsible for implementing automation, AI agent setup, and backend integrations.

Key Functional Needs:

- Seamless parcel booking and tracking interface for customers through Salesforce Experience Site or agent assistance.
- Ability for agents to update parcel status (Booked → In Transit → Out for Delivery → Delivered) in real time.
- Automated notifications and email alerts to customers on every delivery stage update.

- Support for Sender and Receiver details management with address, contact, and email fields for communication.
- AI Agent integration (via Prompt Builder) for real-time parcel status queries and automated assistance.
- Reports and dashboards to monitor parcel flow, delivery performance, and agent activity for admins.
- Security controls with role-based access, profiles, and permission sets for different user levels.

Tools Used:

- Salesforce Developer Org – For configuration, development, and testing.
- Prompt Builder (AgentForce AI) – To create intelligent parcel tracking prompts and responses.
- Salesforce Flow Builder – To automate booking, tracking, and delivery updates.
- Google Sheets / Docs – For initial data planning, requirement documentation, and system mapping.

Identifying Key Salesforce Features & Tools Required

Salesforce Features Planned:

Custom Objects:

- Parcel__c
- Delivery__c
- Sender__c
- Receiver__c

Standard Objects:

- Account
- Contact
- Case
- Task
- EmailMessage
- User

Automations:

- Record-Triggered Flows
- Auto-Launched Flows
- Approval Processes
- Scheduled Flows

Apex:

- Apex Classes for automated parcel operations (e.g., Lead-to-Parcel mapping delivery notifications)

- Batch Apex for periodic updates or notifications

UI:

- Lightning App Page (SwiftShip Tracker App),
- Dynamic Forms for Parcel Details,
- Lightning Record Pages,
- Experience Cloud for Customer Access

AgentForce AI & Prompt Builder:

Integrated AI agent trained to fetch parcel data based on Parcel ID and respond to tracking or delivery status queries

Email Services:

- Custom Email Alerts and Templates for booking confirmations
- Delivery updates
- Completion acknowledgments

Security:

- Profiles,
- Permission Sets
- Role Hierarchies
- Field-Level Security
- Sharing Rules to ensure controlled access for Admin
- Agent
- Customer roles

Designing Data Model and Security Model

Parcel__c (Custom Object)

- Stores core parcel details such as Parcel ID, Status, Weight, and Estimated Delivery Date.
- Linked to Sender__c and Receiver__c objects to fetch sender and receiver details.
- Connected with Delivery__c object to track real-time parcel location and status.

Delivery__c (Custom Object)

- Stores delivery-specific information such as current location, estimated delivery date, and linked parcel.
- Connected with Sender__c and Parcel__c for relationship-based tracking.

- Used by agents to update and monitor delivery status during the parcel lifecycle.

Sender__c (Custom Object)

- Captures sender-related information such as address, contact number, and email.
- Linked to multiple parcels to track sender history and associated deliveries.
- Used for sending booking confirmations and delivery updates via automated emails.

Receiver__c (Custom Object)

- Stores receiver details such as address, contact, and email for communication and delivery confirmation.
- Linked to Sender__c and Parcel__c to associate both ends of the shipment process.
- Used by agents for delivery reference and automated email notifications upon completion.

Security Model Design:

Role Hierarchy:

- Administrator → Delivery Manager → Agent → Customer

Profiles:

- Admin Profile: Full access to all objects and records.
- Delivery Agent Profile: Read/Edit access to Parcel, Delivery, Sender, and Receiver objects.
- Customer Profile: Read-only access to their own parcel details via Experience Cloud.
- Support Staff Profile: Access to Cases and Parcel details for issue resolution.

Permission Sets:

- Used to grant additional privileges (e.g., “Update Delivery Status,” “Access Agent Console”).

Record-Level Security:

- Sharing Rules applied for Parcel__c and Delivery__c to allow controlled access for agents and managers.
- Customers can only view records linked to their own Sender or Receiver details.

Field-Level Security:

- Sensitive fields like Sender Email, Receiver Contact, and Location Coordinates restricted to Admin and Agent profiles only.

Phase-2

Milestone 1- Salesforce Account

Steps Followed

1. Opened the Salesforce website.
2. Selected the Sign Up option.
3. Entered the required details such as name, email address, username, and password.
4. Completed the account registration process.
5. Verified the email address using the verification link received from Salesforce.
6. Logged in to the Salesforce account using the registered credentials.
7. Accessed the Salesforce platform and started working on the project.

Result

The Salesforce account was successfully created and configured. I was able to log in to Salesforce and access the required features for developing and managing my project.

Milestone 2- Objects Creation

To create objects:

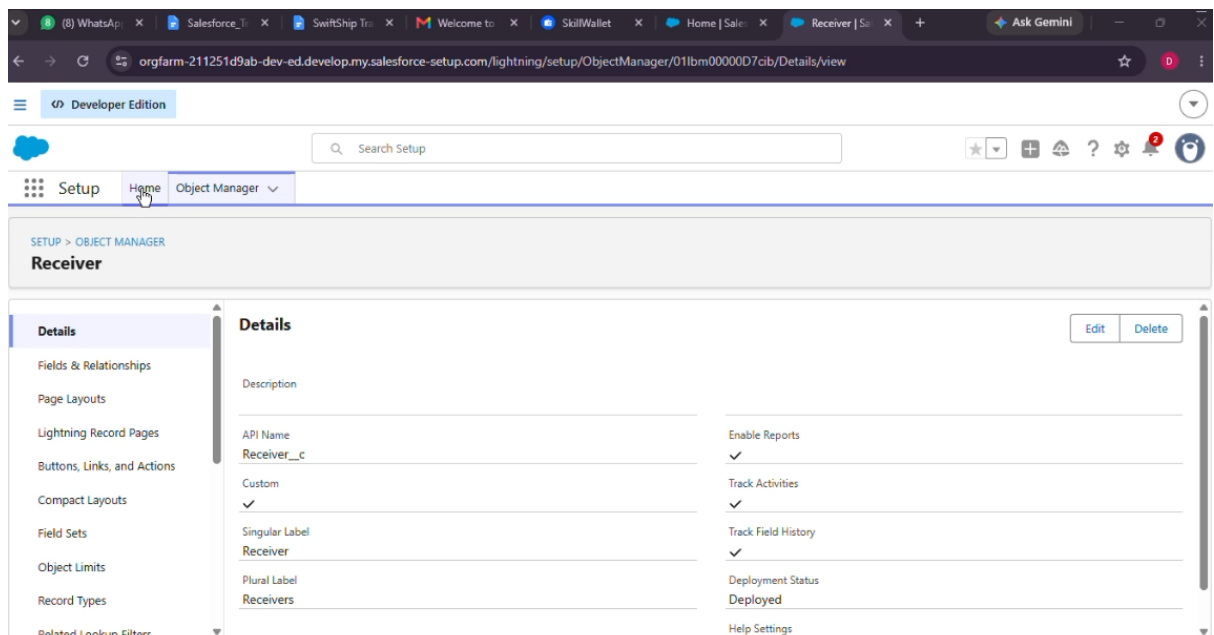
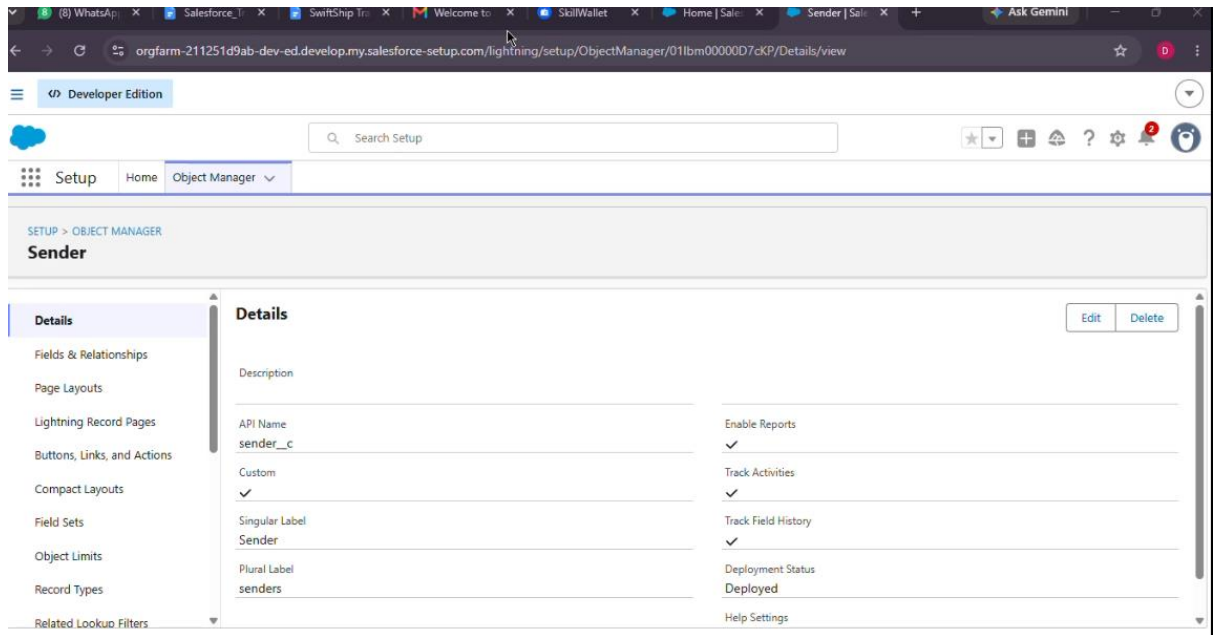
1. From the setup page
2. Click on **Object Manager**
3. Click on **Create >> Click on Custom Object**
4. Enter the label names as **Parcel, Delivery, Sender, Receiver**
5. Enter Plural label names as **Parcels, Deliveries, Senders, Receivers**
6. Enter Record Names as **Parcel Name, Delivery Name, Sender Name, Receiver Name**
7. Select Data Type as **Text: Parcel Name, Delivery Name, Sender Name, Receiver Name**
8. Select **Allow reports**
9. Select **Allow search**

10. Select **Allow Track Field History**

11. Click on **Save and New** for each object.

The screenshot shows the Salesforce Setup interface for the 'Parcel' object. The left sidebar contains a navigation menu with options like 'Details', 'Fields & Relationships', 'Page Layouts', 'Lightning Record Pages', 'Buttons, Links, and Actions', 'Compact Layouts', 'Field Sets', 'Object Limits', 'Record Types', and 'Related Lookups Filters'. The main content area is titled 'Details' and includes a description, API Name (Parcel__c), Custom checkbox (checked), Singular Label (Parcel), Plural Label (Parcels), and a list of settings: 'Enable Reports' (unchecked), 'Track Activities' (checked), 'Track Field History' (checked), 'Deployment Status' (Deployed), and 'Help Settings'. 'Edit' and 'Delete' buttons are visible in the top right corner.

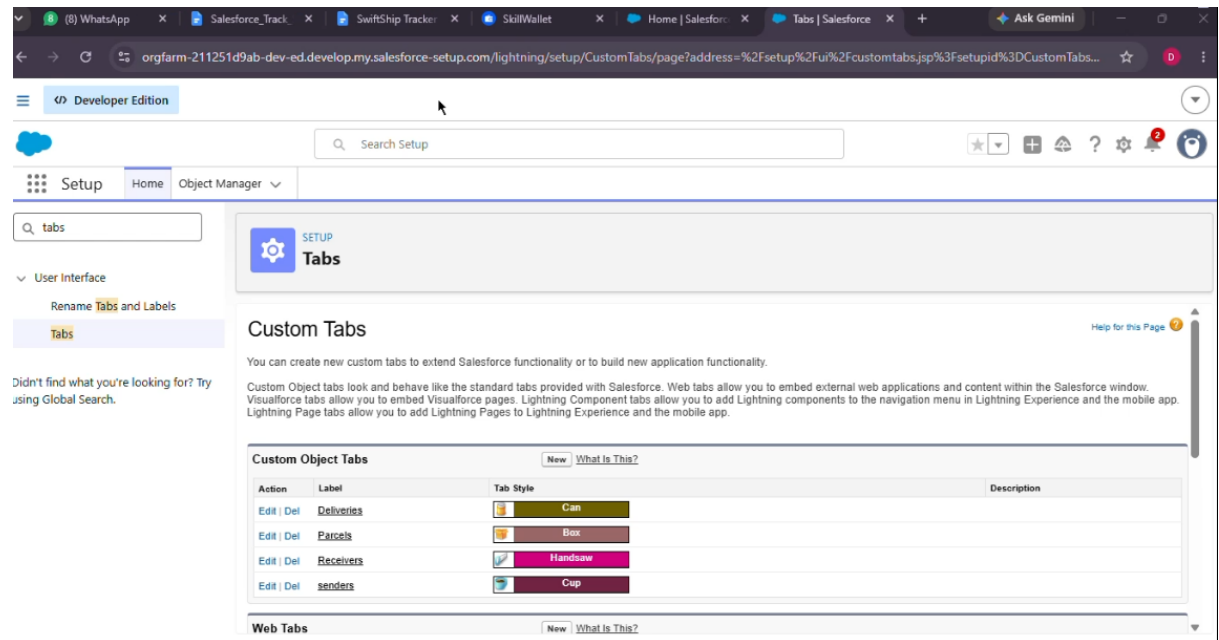
The screenshot shows the Salesforce Setup interface for the 'Delivery' object. The left sidebar contains a navigation menu with options like 'Details', 'Fields & Relationships', 'Page Layouts', 'Lightning Record Pages', 'Buttons, Links, and Actions', 'Compact Layouts', 'Field Sets', 'Object Limits', 'Record Types', and 'Related Lookups Filters'. The main content area is titled 'Details' and includes a description, API Name (Delivery__c), Custom checkbox (checked), Singular Label (Delivery), Plural Label (Deliveries), and a list of settings: 'Enable Reports' (checked), 'Track Activities' (checked), 'Track Field History' (checked), 'Deployment Status' (Deployed), and 'Help Settings'. 'Edit' and 'Delete' buttons are visible in the top right corner.



Milestone 3- Tabs

1. Go to Setup → type Tabs in the Quick Find box.
2. Click Tabs under the User Interface section.
3. Click New under Custom Object Tabs.
4. Select the object (Delivery, Sender, or Receiver) → choose a Tab Style.
5. Click Next → keep the Add to Profiles settings as default → click Next.

6. On the Add to Custom Apps page, uncheck “Include Tab.”
7. Ensure Append tab to the user’s existing personal customizations is checked.
8. Click Save and repeat the steps for each object (Delivery, Sender, Receiver).



Milestone 4- Fields & Relationships

Beside the Quick Find box you have, Object Manager click on it.

Search for and select your Parcel object.

Field 1: Parcel ID

Go to Fields & Relationships → click New

Enter:

Field Label: Parcel ID

Display Format: P-{000} (you can choose your own)

Starting Number: 1

Click Next → Next → Save & New.

3. Follow Step 1 and 2 to create the remaining fields (Status, Weight, Estimated Delivery, Delivery, Sender, Receiver) of the Parcel object mentioned in the table below.

Field Name	Data Type
Status	Picklist
Weight	Number
Estimated Delivery	Date
Delivery	Text
Sender	Text
Receiver	Text

Screenshot of Salesforce Setup - Object Manager for Parcel:

Setup > OBJECT MANAGER
Parcel

Fields & Relationships (9 Items, Sorted by Field Label)

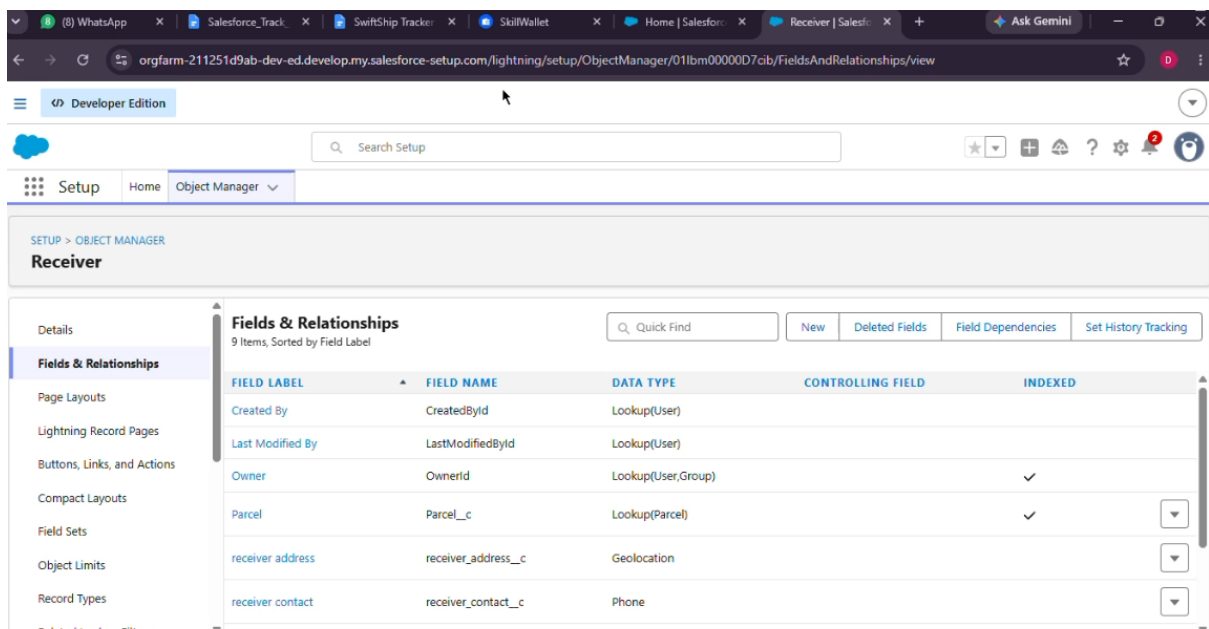
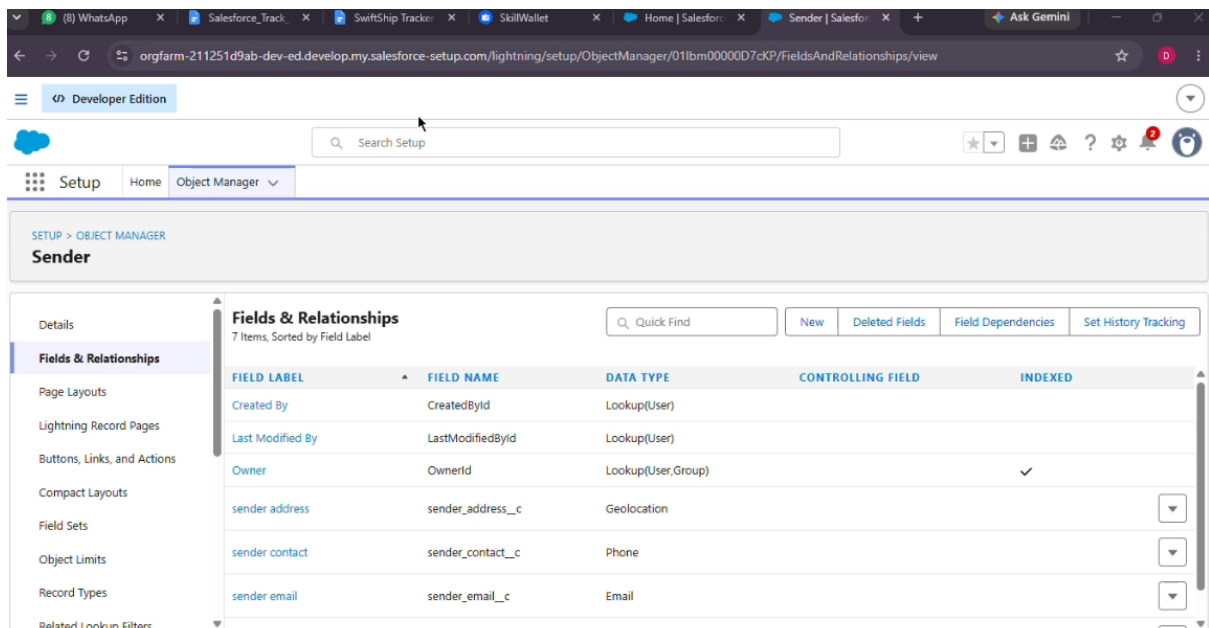
FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Estimated Delivery	Estimated_Delivery__c	Date		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Parcel ID	Parcel_ID__c	Auto Number		
Parcel Name	Name	Text(80)		✓

Screenshot of Salesforce Setup - Object Manager for Delivery:

Setup > OBJECT MANAGER
Delivery

Fields & Relationships (6 Items, Sorted by Field Label)

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
current location	current_location__c	Geolocation		
Delivery Name	Name	Text(80)		✓
Estimated Delivery	Estimated_Delivery__c	Date		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓



Milestone 5 - Prompt Builder

1. In Salesforce, go to Setup.
2. In the Quick Find box, type Prompt Templates (or in Agent Console, go to Agent Builder → Prompt Templates).
3. Click New Prompt Template.
4. Enter the details
5. Label: Retrieve Parcel Details
6. API Name: Retrieve_Parcel_Details
7. Description: Retrieves parcel details where Parcel_ID matches the user input.

8. Enter the Prompt Template Content

Parcel Tracking Update

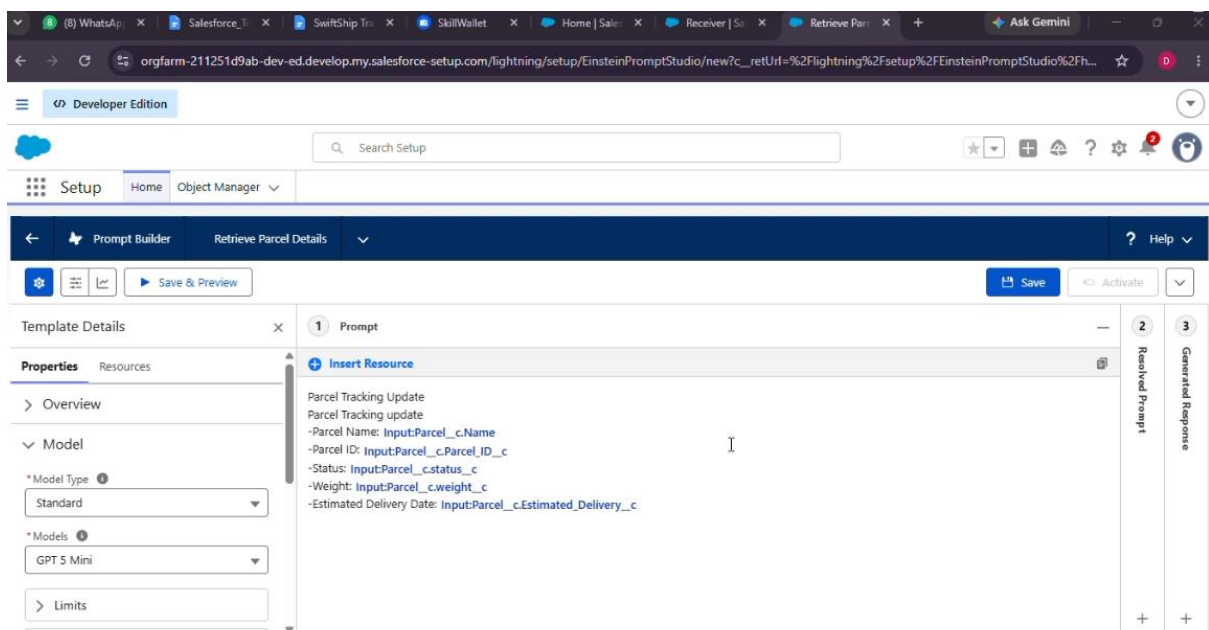
Parcel Tracking Update

- Parcel Name: `{!$Input:Parcel__c.Name}`
- Parcel ID: `{!$Input:Parcel__c.Parcel_ID__c}`
- Status: `{!$Input:Parcel__c.Status__c}`
- Weight: `{!$Input:Parcel__c.Weight__c}`
- Estimated Delivery Date: `{!$Input:Parcel__c.Estimated_Delivery_Date__c}`

9. Click Insert Resource (or Add) for each one.

10. Click Save.

11. Then click Activate to make this Prompt Template available for your agent or flow.

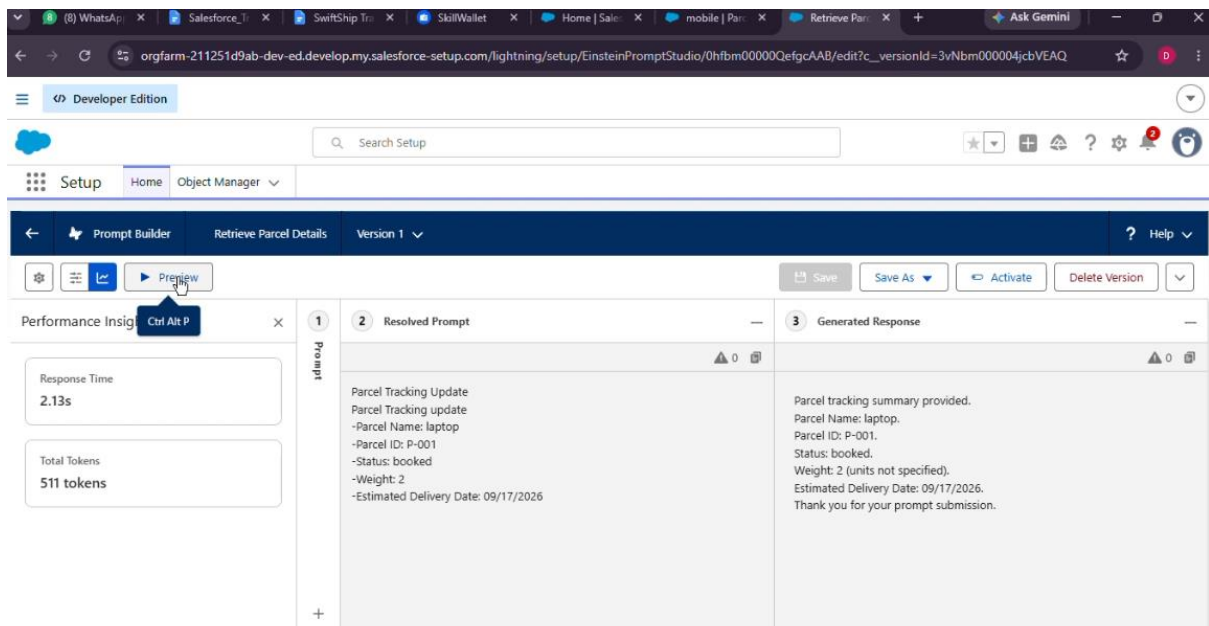


Test the Prompt:

Open your Agent preview.

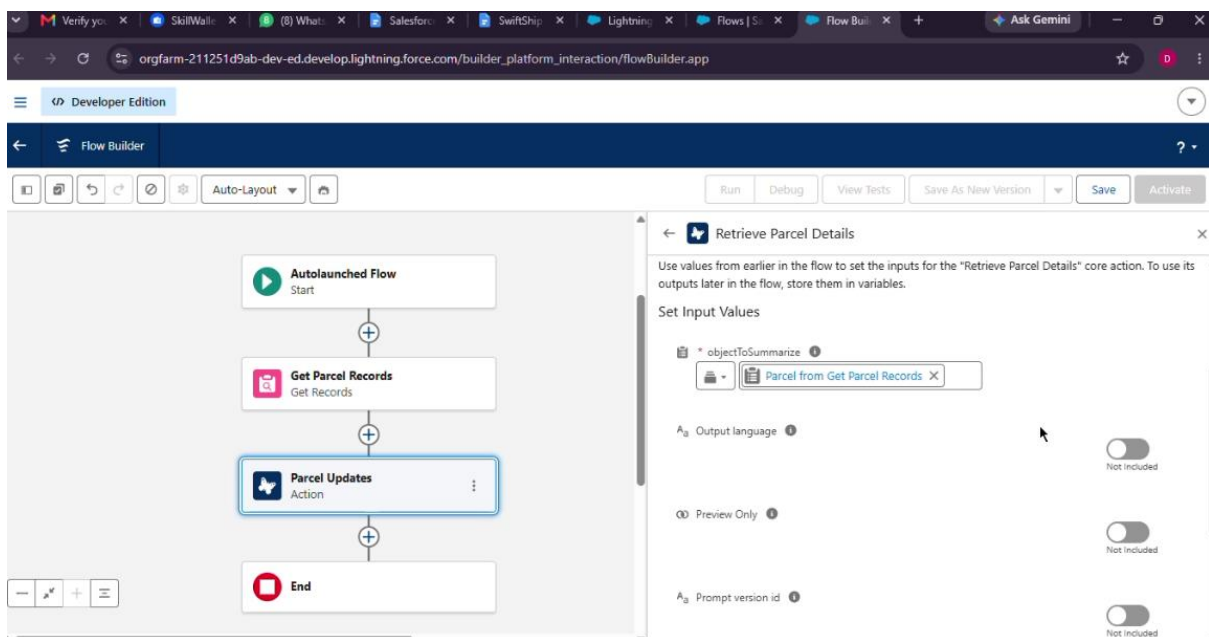
Type: "Track parcel P-001".

It should return details from the Parcel record.

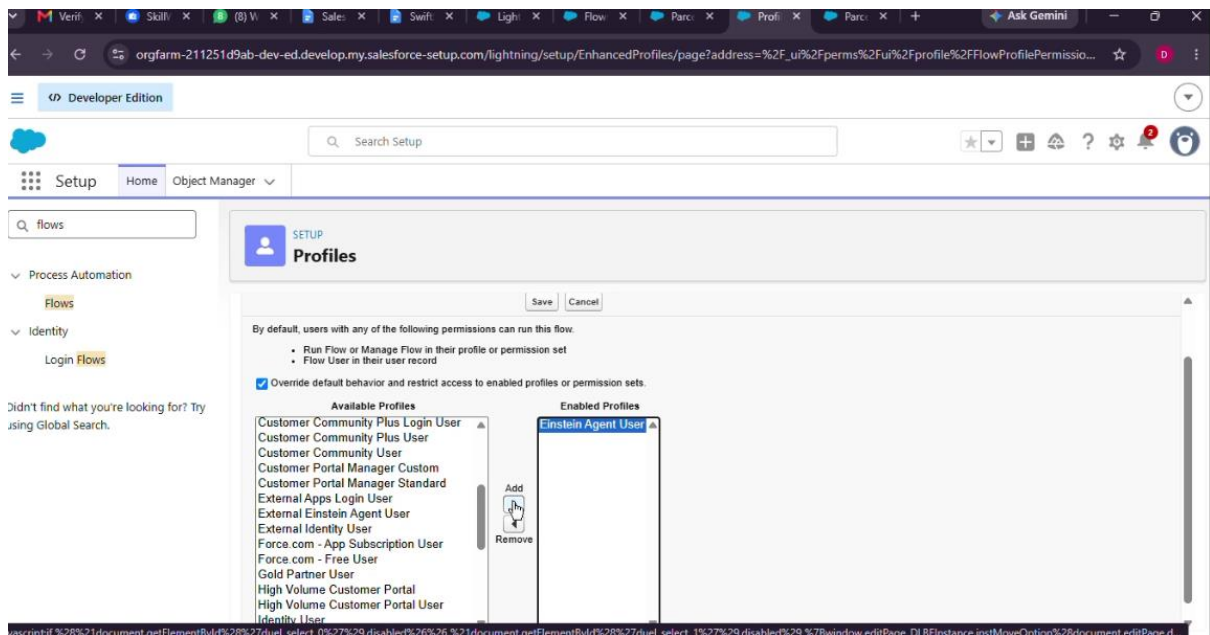


Milestone 6 – Flows

Click on the Add icon to add Element: Get Records→ Label: Get Parcel Records



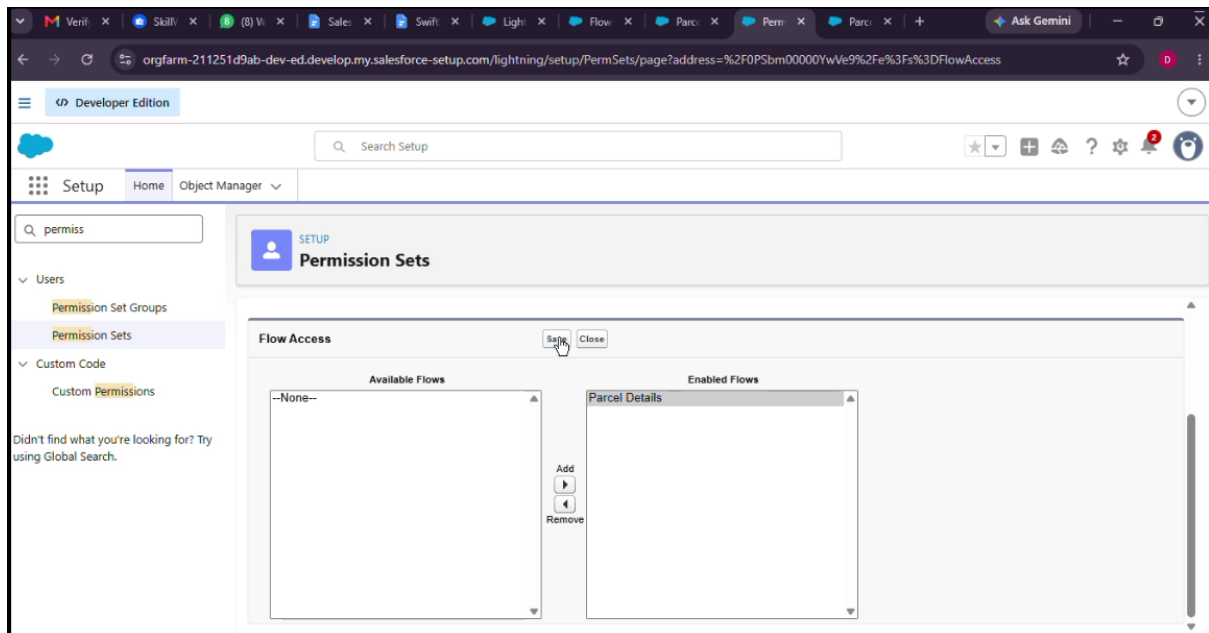
On the top look for flow user access and then assign these available profiles, and click save.



Phase-3

Milestone - 1 Creating Permission Sets

1. In the Quick Find box (left side), type Permission Sets.
2. Click Permission Sets under Users.
3. Click the New button.
4. Enter the following details:
 - Label: Swift Ship
 - API Name: auto-fills
 - License: Choose Einstein Agent.
5. Click Save.



Milestone - 2 Assigning Permissions to your Agent user

1. In the Quick Find box on the left, type Users.
2. Click Users under the Users section.
3. Open the Agent User Record
4. From the list of users, find the Einstein Service Agent user to represent your Agent (e.g., "SwiftShip Agent").
5. Click on the Full Name of that user to open the record.
6. Scroll down to the section named Permission Set Assignments.
7. Click the Edit Assignments button.

→ On the left side (Available Permission Sets), find and select your permission set:

SwiftShip

Click the Add → button to move it to the right-side list (Enabled Permission Sets).

Click Save.

orgfarm-211251d9ab-dev-ed.develop.my.salesforce-setup.com/lightning/setup/PermSets/page?address=%2F0P5bm00000YwVe9%2Fe%3Fs%3DEntityPermissions%26o%3D...

Developer Edition

Setup Home Object Manager

Search Setup

permission

Users

Permission Set Groups

Permission Sets

Custom Code

Custom Permissions

Didn't find what you're looking for? Try using Global Search.

Permission Sets

Shift Ship

Find Settings... Clone Delete Edit Properties Manage Assignments View Summary

Permission Set Overview Object Settings **Parcels**

Parcels Save Cancel

Tab Settings

Available	Visible
☐	☐

Object Permissions

Permission Name	Enabled
Read	☒
Create	☒
Edit	☒

orgfarm-211251d9ab-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ObjectManager/011bm00000D7c2f/FieldsAndRelationships/view

Developer Edition

Setup Home Object Manager

Search Setup

SETUP > OBJECT MANAGER

Delivery

Details

Fields & Relationships

6 Items, Sorted by Field Label

Quick Find New Deleted Fields Field Dependencies Set History Tracking

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
current location	current_location__c	Geolocation		
Delivery Name	Name	Text(80)		☒
Estimated Delivery	Estimated_Delivery__c	Date		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		☒

The screenshot shows the Salesforce Setup interface for the 'Sender' object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, and Related Lookup Filters. The main content area is titled 'Fields & Relationships' and shows 7 items, sorted by Field Label. The table below lists the fields:

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
sender address	sender_address__c	Geolocation		
sender contact	sender_contact__c	Phone		
sender email	sender_email__c	Email		

The screenshot shows the Salesforce Setup interface for the 'Receiver' object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, and Related Lookup Filters. The main content area is titled 'Fields & Relationships' and shows 9 items, sorted by Field Label. The table below lists the fields:

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Parcel	Parcel__c	Lookup(Parcel)		✓
receiver address	receiver_address__c	Geolocation		
receiver contact	receiver_contact__c	Phone		

Milestone -3 Org Setup for AgentForce

1. Go to Setup → Quick Find → Einstein Setup.
2. Open Einstein Setup → click Get Started if prompted.
3. Toggle On these key services:
4. Turn On Einstein
5. Global Languages
6. Deploy Prompt Template
7. Refresh your browser page.
8. Go to Setup → Quick Find → AgentForce Agents.

9.Toggle On these key services:

Agentforce

Enable Agentforce Default Agent

Milestone - 4 Building The Agent

1.In Setup, search for Agentforce Agents in Quick Find.

2.Click Agentforce Agents, then click on the +New Agent, Then click Create Agent

3.Then create the agent

Label: SwiftShip Tracker

Developer Name : Auto fills

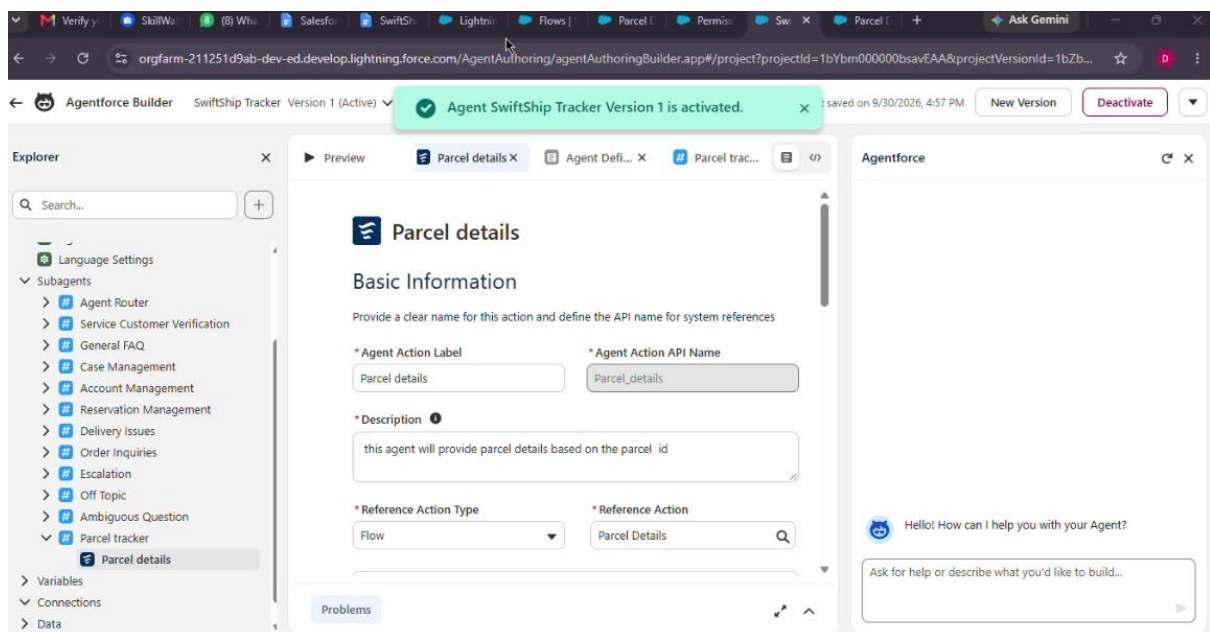
4.Click on New Agent Action.

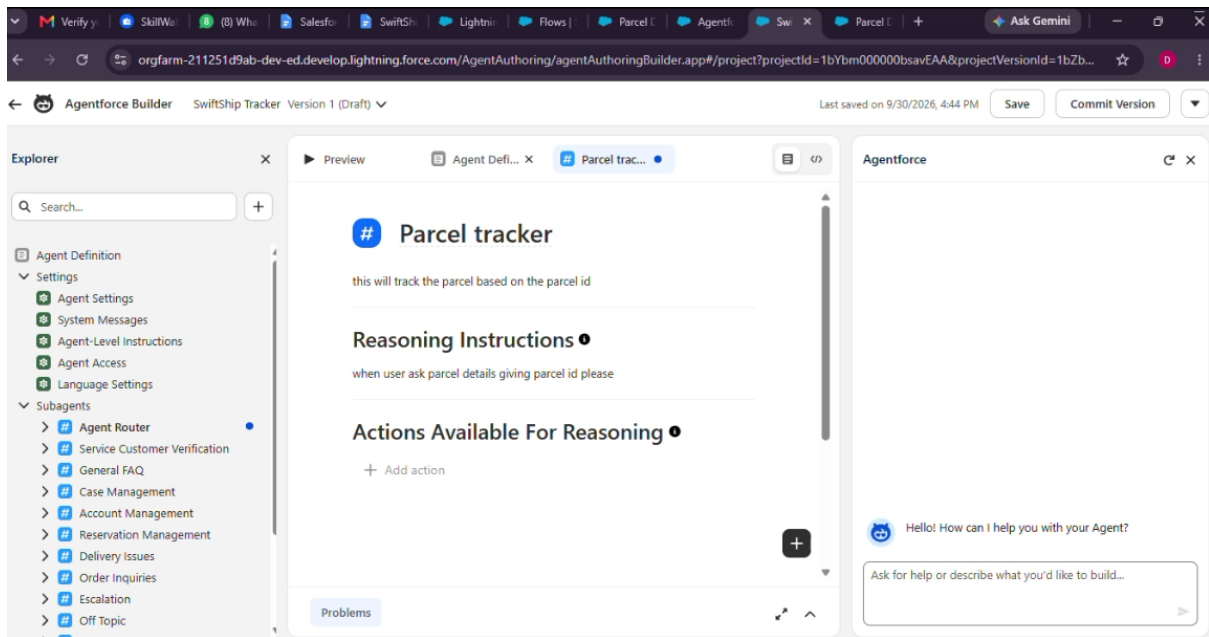
5.For Action type, Click on Flow then Next.

6.For Reference Flow use the recently created flow, click Next.

7.Fill the details for Agent Configurations

8.Click Save.





Milestone - 5 Assigning Permissions to your Agent user

1. In the Quick Find box on the left, type Permission set.
2. Click Letter S
3. Find Swift Ship Permission set
4. Click Shift Ship.1
5. Click Manage Assignments
6. Click Add assignments
7. Check EinsteinAgentUser
 - Assign and Done
 - Come Back to the Agent SwiftShip Tracker
8. Click on Commit Version and Activate, then Activate.

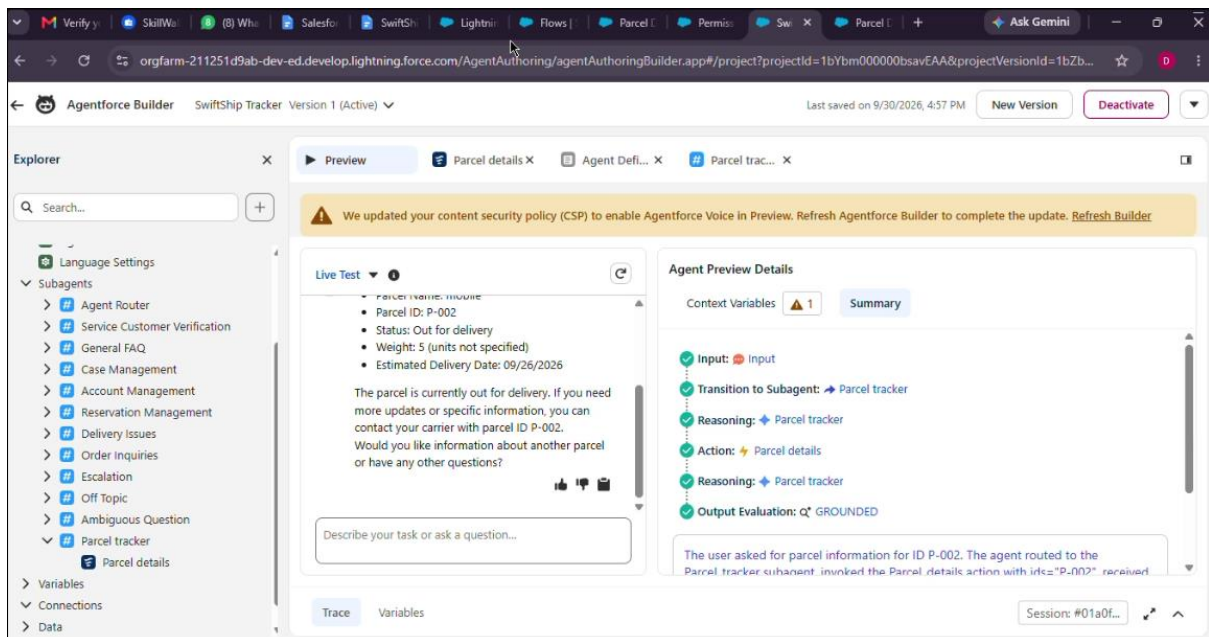
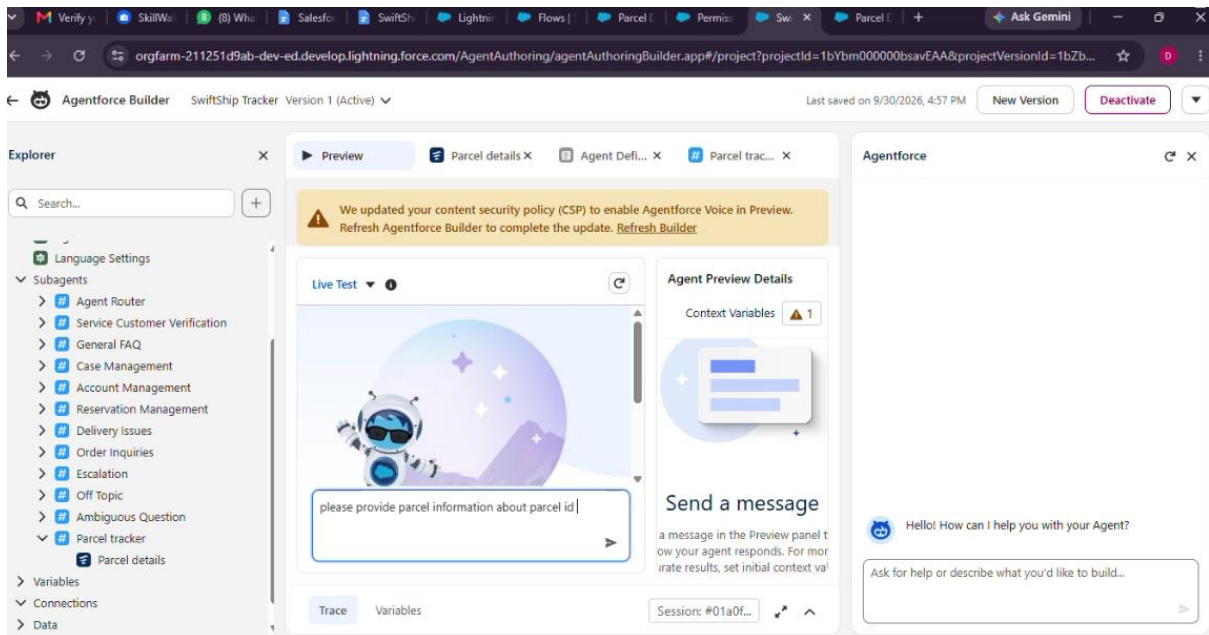
Phase-4

Data Migration, Testing & Security

1. Go to Setup → Agent Builder.
2. Open the Agents list and confirm that your agent Name: Parcel Tracking Agent, appears.
 - Check details:
 - Description: "Helps customers book, track, and manage parcel

Type: Service Agent

Default Agent: Confirm if toggled ON (so it responds automatically)



Phase 5: Deployment, Documentation & Maintenance

Project Demo Video Presentation

https://drive.google.com/file/d/1BHt8WBcz_bWp1RtlkFHpu4OOe53TvME7/view?usp=sharing

Conclusion

The Salesforce Parcel Management project was successfully developed to manage parcel details efficiently. Custom objects and fields were created to store important information such as Parcel ID, Status, Weight, Estimated Delivery, Delivery, Sender, and Receiver. This project helped in understanding Salesforce Object Manager, custom objects, fields, and data management. Overall, the project provides a simple and organized way to manage parcel information using Salesforce.